Designing Our Way Through Abstractions:

Calling for More Practice-based More-than-Human Design Research

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ABSTRACT

Human-computer interaction (HCI) researchers increasingly recognize the importance of attending to other-than-humans in their practice. The More-than-Human (MtH) turn to HCI, seeking to challenge human-centered approaches, is undergoing rapid evolution with exciting advances on many fronts. Yet, abstract notions of MtH theories become quickly messy in practice. Here, we emphasize the need for practice-based MtH design research to unpack these. We discuss five case studies of our own work to illustrate how MtH research can be performed from the bottom up, leaning heavily on hands-on design practice (rather than solely theory-driven reasoning): backpacking in nature, growing mushroom composites, performing with robots as nature companions, walking in a city to attune gravel and geological processes, and playing nature in video games in MtH design contexts. Our contribution, thus, foregrounds the value of embodied, situated, and designerly approaches, encouraging research that makes MtH design actionable from a practical perspective.

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CSS Concepts

• Human-centered computing ~Interaction design ~Interaction design theory, concepts and paradigms

INTRODUCTION

The traditional human-centered approach to design is increasingly being challenged by advocates of a More-than-Human (MtH) perspective. This approach moves beyond human-centered design towards an approach that decenters humans, acknowledging the intertwined influences and needs of MtH actors in the design [39]. However, despite its theoretical promise, translating MtH concepts into hands-on design work presents significant challenges. Abstract ideas, such as decentering humans in design, often become ambiguous and messy in practice [30]. Thus, it is crucial to shift MtH design from a predominantly theoretical discipline to one that increasingly explores what it means to engage with the MtH in practice [9]. Emerging research in this area explores how practice-based MtH design can actively engage with and make sense of the abstract ideas of MtH through hands-on experiences and reflective dialogues, e.g., [6, 7]. These works, whether they depart from theoretical reasoning or not, emphasise the value of design action and reflections, contributing new insights into what MtH design could be in action.

In this pictorial, we aim to further the body of practice-oriented approaches in MtH design research by looking at a collection of cases where we produced MtH design knowledge through active design practice (rather than solely theory-driven reasoning). Here, we base our work on design being an inherently knowledge-creating discipline, not just an application of existing theories [32]. We frame practice-based MtH design research as an action-oriented practice capable of generating MtH insights, aligning with Schön's [34] concept of design as action-reflection cycles. Our approach also acknowledges that the design actions and knowledges created through them are influenced by personal experiences, existing theories, tools, society, and culture, making design a non-neutral activity [22].

Aligned with this, we aim to highlight practice-based MtH design research as a way to produce insights into MtH design: We examine the ways in which prioritizing practice contributes knowledges related to discovering our relationships, dependencies, and interconnections to broader MtH worlds in the context of design. For this, we present a compilation of first-hand design research instances from the Gamification research group at Tampere University, covering a wide range of actions, contexts, and other-than-human engagements. By doing so, we contribute MtH design research insights that are produced in a radically practice-based, driven by (design) action rather than by theoretical reasoning.

TROUBLING ABSTRACTIONS IN MTH DESIGN

The MtH concept advocates for a broader perspective that includes other-than-humans. This perspective rejects the Enlightenment view of separations

between, e.g., "subjects and objects, nature and society" [10]. Instead, it adopts a hybrid approach that views humans and other-than-humans as part of interconnected systems [41]. Central to this concept is posthumanism, which promotes "nonexploitative forms of togetherness" rooted in relationality and pluralism [25, 26]. This approach requires a shift from traditional human-centered design to one that acknowledges the intertwined influences and needs of MtH actors in the design [39]. However, the implementation of these theories in practical design processes poses challenges. Abstract concepts derived from theory quickly become ambiguous and messy when applied in real-world contexts. For example, investigating the notion of "decentering humans" in posthuman theory and MtH design practice, Nicenboim et al. [30] unpack the gap between theory and practice. Their review identifies how practice poses varying interpretations of the concept that are often not explicitly connected or reflect the theories related to decentering. Nevertheless, these practices, whether they depart from theories or not, produce novel knowledges in the form of methods, and practical insights relevant to MtH. This underscores the need for more practice-based MtH design research, moving beyond theoretical discussions [9]– a need we address in this pictorial.

We believe the practical understanding of MtH in design is crucial because, as we see it, focusing solely on MtH theories might lead to similar criticisms that theories about animal rights face [28]: They present abstract concepts for better addressing complex issues, but their connection to the real-world is messy and requires further research. By referring to Haraway's "staying with the trouble" [23] as an inspirational point of departure, in this pictorial, we invite researchers to acknowledge the practice-based approaches' power in revealing not only the messiness of MtH theories in action but also their potential to reshape our current knowledges.

RELATED WORK

There is a growing body of research presenting practice-based approaches for MtH design research: For example, Karana et al. reflect on design practices with biomaterials, highlighting insights such as their unique temporalities and potential for mutualistic care relationships [24]. Behzad et al. focus on their experience of designing Morse Things, ceramic artifacts connected to the internet, emphasizing how attention to a other-than-human agency, breakage, can contribute to design processes [6]. By deploying Morse Things in households and examining MtH relations, they also uncovered MtH relations, e.g., sharing experiences with the designers, researcher through things or humans attuning to things' ways of communication [38]. Dolejšová et al. [12] eport insights into MtH food practices through a series of hands-on practices such as embodied, improvisational engagements with ingredients and cooking equipment. Søndergaard and Woytuk [35], through participatory crafting and reflections through biomaterial-based care practices and artifacts, provide insights into how MtH bodies can be practiced. Dörrenbächer et al. use roleplaying methods to approximate human experiences to those of robots, providing methodological insights into understanding other-than-human perspectives [14]. Reflecting on designing interactive animal feeding stations, Brown et al. [7] provide concrete instances of challenges and strategies for designing both humans and other-than-humans.

Altogether, these approaches demonstrate how situated experiences can generate meaningful insights into and foster new understanding of MtH in design practice. We believe this practical approach to MtH through design is crucial and requires more attention to be paid. Contributing to this, in the next sections, we take a closer look at how and what kind of new insights related to MtH are achieved in our research, in ways that are radically situated, based on (design) action.

PRACTICE-BASED MTH DESIGN RESEARCH

Before introducing our framing of practice-based MtH design research, given the influence of ourselves in this inquiry, we first outline our positionalities: We are HCI researchers based in Finland with diverse origins—Türkiye (Çağlar), Spain (Ferran), Bangladesh (Esthiak), Lithuania (Linas), and Germany (Velvet). Our research interests span across the spectrum of MtH design, from understanding joyful and caring (Ferran), playful and caring (Velvet) and companionship (Esthiak) relationalities to discovering the other-than-human influences in and for design practices (Ferran, Çağlar and Linas). Despite these differences, we all share the belief humans are not separate from other-than-humans neither in existence nor in design processes: We are entitled to discover these interconnections and dependencies in MtH worlds for finding ways to co-exist rather than exploit the other-than-humans for human benefits. From this perspective, we focus on practice-based MtH design research as opportunity to explore this belief in practice:

We see design as an inherently knowledge-creating discipline, not just an application of existing theories [32]: the MtH designer produces ***knowledges*** about the ways to co-exist in MtH worlds in practice through ***actions*** taken with other-than-humans (such as living or crafting together) and by ***reflecting*** on these embodied experiences (Figure 1). This framing is heavily influenced by Schön's framing of "Reflective Practitioner" [34] who describes design as a reflective dialogue between designers and design situations during and after their design actions. Yet, this is not to say that the practice-based MtH design research is independent of theories. Instead, we acknowledge that neither our design actions nor our reflections are neutral [22], it is situated, ***mediated*** through not only theories that we know of but also our previous experiences, values, tools, the other-than-humans that encounter, our cultures, and societies at large, hence our positionalities. The value of action in practice-based MtH design comes from its power to situate ourselves in MtH contexts, encounter other-than-humans and reflect on our experiences to produce knowledges through first-person, embodied, relational and phenomenologically situated manner [18].

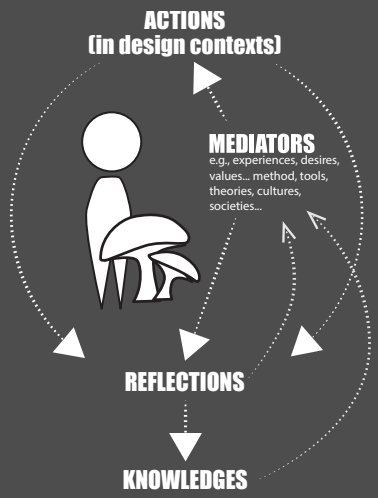


Fig. 1 *Our framing of Practice-based MtH Design Research illustrated.*

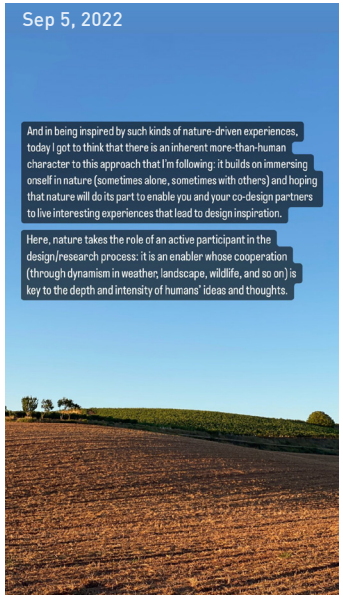
In the following pages, we document and reflect on five such first-person experiences and describe the unique ways that embodied practice-based activities contributed to our understanding MtH dependencies and relationships within our design contexts.

CO-DESIGNING FROM-THE-WILD THROUGH HIKING/BACKPACKING:

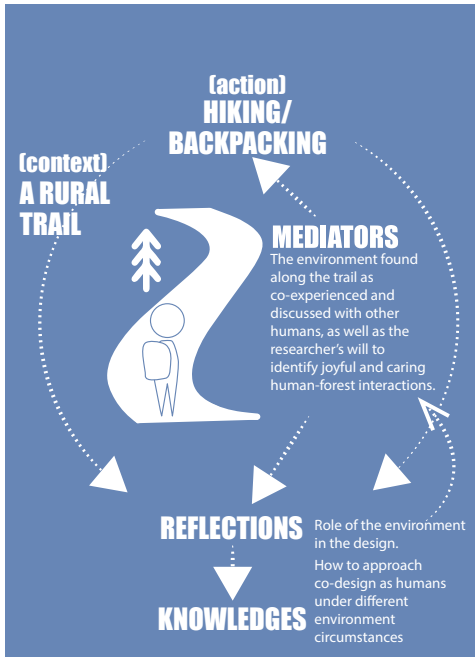
(FERRAN)

As part of my research into how to design technology that supports joyful and caring human-forest interactions, in 2022 I did a study where I immersed myself in a one-month backpacking adventure. The aim of the study was to co-experience nature with other backpackers as a platform for engaging them in generative [19] discussions about how technology might contribute to realizing the joyful and caring potential of those experiences. Through the study, I also sought to further develop, in a hands-on and situated manner, the from-the-wild [2] methodology our group has begun to formulate, i.e. an approach to interaction design and research that builds on the idea that design research concerned with nature should be enacted from the very nature, engaging skin to skin with both humans and non-, rather from a human-made environment.

Over 800 km+ hiked for 30 days, I co-experienced a broad range of landscapes and interacted, in different depths and lengths, with 200+ backpackers from 35+ nationalities to co-imagine joyful forest tech futures with them. A rich documentation of the trip can be accessed on the project's Instagram (@wildtechresearch), and slices of that project have been and will be disseminated in other publications (e.g., [1, 3, 4]). Here I focus on a specific set of events within that study, where engaging hands-on with from-the-wild activity gave rise to a set of methodological reflections on the role of other-than-humans in design activity targeting and taking place within nature.



"The situation I'm focusing on was captured through Instagram stories, which I kept producing on a daily basis throughout the backpacking study to capture the key reflections and ideas produced each day. The reflections I share here emerged during the hikes of two specific days; they evolved and were subsequently synthesized into a set of Instagram stories at the end of each of those days. Weeks after the backpacking trip concluded, I did a second-wave reflection on those materials, which led to an extended reflection in textual form."



REFLECTIONS

By experiencing first-hand what it means to co-design from the forest, I'm reflecting on what's the potential and what are the implications of that move. For example, I've realized that in this kind of work, the forest is not only an object of research, or a habitat within which the research takes place: it's, in some way, an active participant in the research process. The way it works is: you immerse yourself in nature, let it be with others or not, and then wait for nature to do its thing, to "throw stuff at you" so you can live interesting experiences and translate them into design inspiration. If the forest doesn't cooperate, if no interesting events take place, there's not much research to be done, really. For example, if there's a very similar and stable weather day after day, if the landscape is really monotonous, if there's not much wildlife around… During my backpacking experiment, some days were like that, and I can assure you that the conversations and ideation suffered from it. So it turns out that, in this kind of work, nature may be a direct and active participant of the design and research process after all: if it doesn't cooperate, the depth and intensity of thinking will decrease. One could therefore say that this kind of approach has an inherent MtH twist to it, as it involves nature as a crucial stakeholder in the design process.

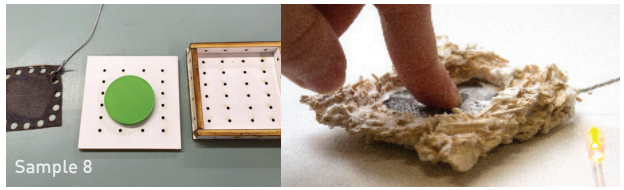
What's interesting about this example is how, by engaging with design practice in a radically situated and hands-on way, a set of exciting reflections on MtH design research methodology emerged: (1) ***When you co-design from-the-wild, other-than-humans inevitably play a role as enablers: they provide stimuli and enable experiences that will become the inspirational material for humans to design with.*** (2) ***The absence of such other-than-human stimuli will likely compromise our ability to ideate.*** (3) ***In the event of such absence, one might want to focus on higher-level reflection or meaning-making, seizing the absence of stimuli as an opportunity to temporarily detach from context.***

GROWING MYCELIUM COMPOSITES:
(ÇAĞLAR)

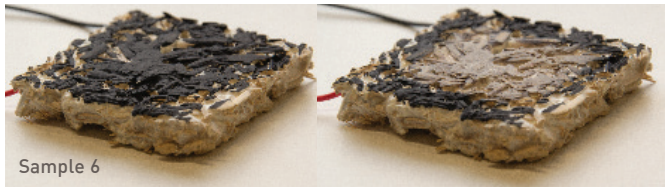
This research presents material experimentation with mycelium, the vegetative part of the mushroom, together with other interactive materials, such as conductive yarns, LEDs, thermochromic dyes, heating units, and microprocessors. The goal was to combine these materials to explore design opportunities around interactive mycelium composites by also understanding active influences of mycelium in shaping the designs. Detailed in another publication [20], the design process involved a series of experimentations with growing mycelium over the course of 6 months, conducted by me with the help of my co-authors from the aforementioned publication: I first attempted to grow mycelium in containers to observe and understand how mycelium grows. The second step was then to create conceptual designs based on my insights. At this stage, I sketched out what kind of interactive composites we wanted to experiment with and planned out how to combine all materials. Finally, the final stage included producing a set of 3D printed molds, putting mycelium growth medium together with other materials into these molds and letting them grow.

TURN-TAKING IN MTH DESIGN

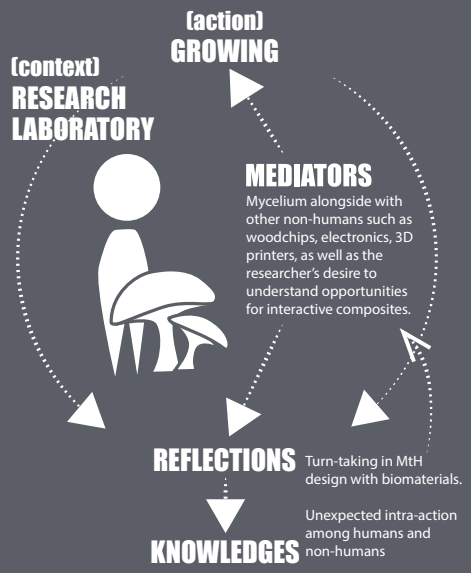
"The slow growth of mushrooms, attuning their temporality, required me to design first, provide an environment for mushroom to grow, wait for mushrooms to do their thing. When they grow enough, I observed, touched, understood them and, then, re-design, initiating a new cycle of turn-



"One of the unexpected instance was when the mycelium grew over the conductive fabric. This did not only defined the final form but it also impacted the functionality, hindering the conductive touch from time to time"



"The color-change was defined by how much mushroom covered the spaces between wood chips. It impacted how far the thermochromic dye is from the heating unit, varying the speed of color change. At the end, it was the interplay between my ideas, woodchips, mushrooms, thermochromic dye (among others) that formed the interactive pattern."



REFLECTIONS

The first thing that I found fascinating about my growing interactive mycelium composite experience was ***the possibility of turn-taking with the growth of mycelium in design***. The mycelium growth is relatively slow when I compared it to e.g., my experience of immediately shaping a wood piece for designing a chair. This forced me into an iterative cycle of co-designing where we took turns with mycelium in design: I took my turn to design and provide an environment for the mycelium growth and mycelium took its turn to grow in unique ways, giving me instances of how I can take my turn and do the first step again.

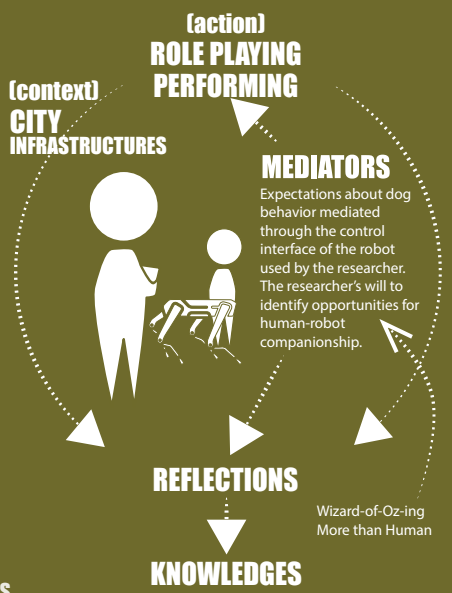
The second takeaway was that it was ***the unexpected intra-actions among me and many non-human actors (mycelium, wood chips, electronics, 3D printed molds, and so on), that collectively defined interactivities and the look of the composites***: To name some concrete instances, the mycelium and electronics interacted in the ways that I did not anticipate: For instance, in sample 8 (S8), I designed a mold for blocking the mycelium's growth on the part where I placed the conductive fabric. Yet, unexpectedly the mycelium grew over some parts of the fabric (and the fabric let it to do so), yielding a blended look between these actors and making it harder to detect conductive touch on some parts of it. Similarly, while initiating the growth, I placed a heating (peltier) module within a block of mycelium growth medium, on which I applied a thermochromic dye after its growth. While I expected when I heat up the peltier module the color of the thermochromic dye would change, in the resulting composite this change varied depending on how close the dye was to the heating module which was defined by the mycelium growth.

PERFORMING
ROBOTS AS NATURE COMPANIONS:
(ESHTIAK)

My research investigates how robots can evolve into genuine companions for humans and explores how agency can be developed in robots to make them feel like companions rather than mere command-following entities. Additionally, it delves into designing robots that can co-exist and co-perform with humans in natural environments such as forests or the wild, emphasizing intelligence, agency, and rationality. This involves moving beyond human-centered design to incorporate more-than-human perspectives, including the roles and agency of nature and its elements in human-robot interactions.

To understand the design space of companion robots in the wild, I conducted a Wizard of Oz study where I investigated how humans perceive recreational walking accompanied by a robot dog. During this study, I used an interface to control a robot. As the controller of the robot, I had to contextually think and perform as a dog would do. I constantly had to think about how a dog would behave against the human's interactive cues. In other words, I embodied the robot dog and acted as the dog, painting a somewhat relatable experience for me to understand what a robot or dog perspective could be like.

"Seeing the human and the robot from the wizard-of-oz perspective. As the controller of the robot, I had to look at the human, see how they try to interact with the robot as well as the environment, and then channel my embodiment of the robot to the controller. I was constantly thinking how the robot dog should behave in each situation."



REFLECTIONS

My experience of controlling the robot in a Wizard-of-Oz study offered a profound glimpse into MtH design perspectives: By embodying the robot dog and responding contextually to human interactive cues, I experienced firsthand the complexities of creating a believable and empathetic non-human companion. I constantly thought about the expectations and norms regarding what it means to embody a robot dog and asked myself questions such as "What are the expectations from a dog itself?", "What is too much like a dog? How to balance the doglikeness? How is the doglikeness mediated?" I tried to balance what I thought were human's expectations of me as a robot dog and negotiated with robot's capabilities to comply with them. Thinking about how a real dog would contextually behave and expressing it through the affordances of the robot dog was somewhat restricted at times because of the limitations of the robot. This also indicates how mediators (the robot dog in this case) of MtH practices might influence the overall interaction and perceived experiences.

Overall, my experience of controlling a robot dog has been an effective way to understand their perspectives. It inspired me to think about and create ways to embody non-human and planetary objects so that we understand them better. The wizard-of-oz method in this case worked effectively to prompt thinking about how a robot dog would approach recreational walking with a human.

WALKING THE GRAVEL CITY:
(LINAS)

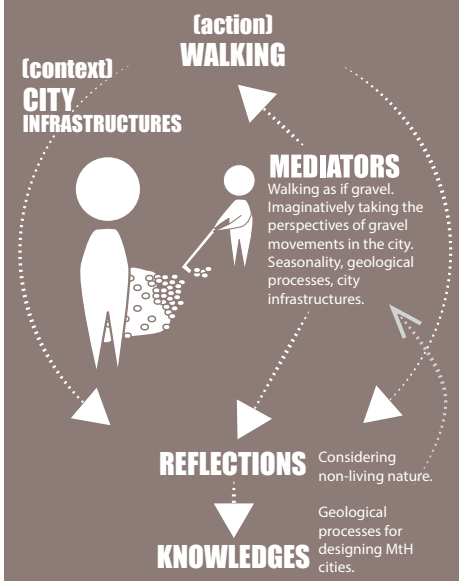
My research aims at fostering a personal and playful relationship with the geology of Tampere, Finland, I performed a range of walking activities to experience the entangled connections between human infrastructures and the life of gravel in the MtH city. Uniquely positioned on one of the world's tallest eskers, Tampere is on a hill, made out of gravel left behind by melting glaciers. A particularly interesting aspect of the geological interplay of the city is the use of gravel to manage the city's winter conditions. During the snowy months, gravel is spread across roads and pathways to provide traction and prevent slipping. As winter transitions to spring and the snow begins to melt, the gravel that was spread across the city accumulates into a series of naturally artificial formations- due to meltwater runoff, street cleaning, urban movements, and other activities. After the season changes and all of the snow melts, the city then re-collects all the gravel from the streets, where it is cleaned and prepared for the next season.

Previous studies have highlighted the importance of soil and ground in MtH cities [33], and have acknowledged the importance of movement and dynamics in understanding the solid grounds on which humans are positioned [8]. Parallel efforts prioritize movement and movement ecologies as central to understanding dynamic, relational worlds [29], or employ relational perspectives in understanding natural resources [5]. As HCI and design aim to involve geological deep-time perspectives in their toolkits, postphenomenological human-world relations can be expanded into human-technology-geology relations [31]. To explore these perspectives as potential avenues for design activities, I draw on MtH walking methodologies [36], with a focus on experimental artistic acts of human-geological walking as a way of attending to entangled, site-specific affects, atmospheres and boundaries of 'stone-y urban environments' [11, 13]. My research focuses on walking the Tampere as a gravel-city to examine how geological materials, particularly gravel, move through and with MtH environments of the city. The gravel trails were followed over a month period, during the season transition period. Pictures were captured and reflected on over the course of several days.

"When the snow melts away, gravel emerges as a slow, yet somewhat active participant in the city's movements. As I walk, gravel gets pushed around, sometimes bouncing away from under the pressure of my shoes. Sometimes, the gravel gets caught in my shoes, and sometimes it escapes, lodging itself in doorways preventing them from closing properly. As I consider the life of gravel through its movements in the city, it also makes me imagine the deep-time journey gravel has taken to arrive here."

"The remains of melting snow, infused with gravel, accumulating in piles. The piles are formed through a collective effort by the city services that shovel the streets from snow, as well as the services that collect and clean the gravel from the streets as the season changes."

"As I walk along the city streets, I notice how gravel accumulates in the narrow spaces between water drainage paths and tire tracks left by passing cars. Guided by the melting snow, these tiny stones create winding trails that map the flow of water and the movement of vehicles. Each step I take follows these gravel paths, revealing the subtle interactions between natural forces and urban life, where every pebble's journey tells a story of gradual, persistent change."



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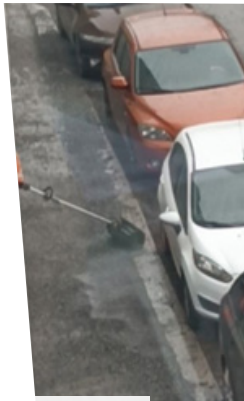
Geological processes for designing MtH cities: My experience of walking-as-gravel led me to reflect on the non-organic forms of life in the city. Cities encompass not only humans and other living entities but also non-living actors like gravel. Walking through the gravel-filled city allowed me to observe the life of gravel during changing seasons, where it seemed to act 'alive' through its movements and flows. This experience encouraged a geological reflection on MtH perspectives. Although I was tempted to describe gravel in terms of liveliness, vitality, and agency, my commitment to the geological allowed me to attune to the interplay between living and non-living material flows within the city.

Following gravel trails allowed me to prioritize gravel's movement ecologies, making space to experience changing seasons through a MtH perspective. By attuning to gravel, I recognized how the city's processes mirror and extend geological phenomena such as sedimentation and erosion. The seasonal movement and accumulation of gravel are tangible reflections of ongoing geological processes. Thinking about mobile city terrains through the slow movements of gravel around the city highlighted these dynamics. This practical activity helped me reflect on the active influences and intra-actions among the technological and infrastructural processes of the city. This reflection on the city's movement ecologies led me to consider how urban design entangles human movements with non-organic elements, positioning the city as an extension of Earth's processes.

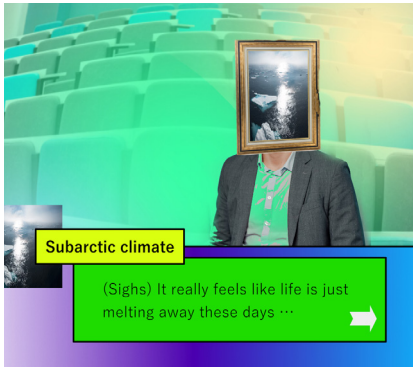


"The city employs state workers with specialized tools to unstick the gravel from the ground. Noticing gravel also made me notice the tools and processes through which the city facilitates gravel movement in the city."

PLAYING NATURE IN VIDEO GAMES:
(VELVET)

I undertook an auto-ethnographic study in which I played video games that prominently feature nature in them, in 2021. As first-person research [16] , I sought to understand how the portrayals of digital nature would respond to my own sensibilities (as a sensitizing activity), and if I could affectively attune them (after Tsing [37]). To offset my (negative) experiences of playing with nature in video games, I created several sketches of video games (presented here) that explore a different set of interactions and narratives than the ones that I experienced.

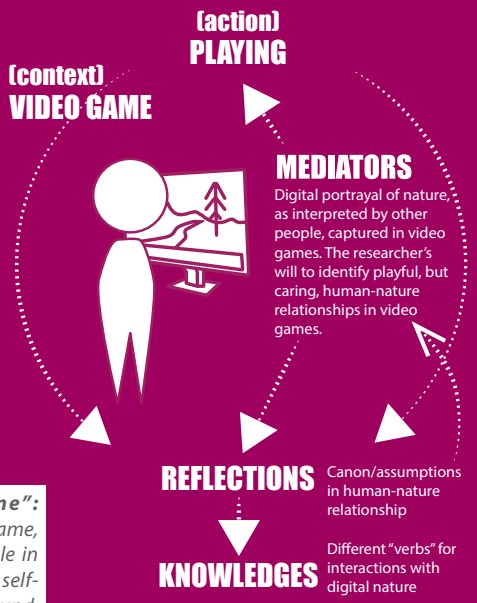
"The body is a vessel": *In an exploration-focused walking simulator, you play as a humanoid protagonist. You are given a choice of elements, objects and nutrients to ingest. Then, after having made your choice, you are free to travel around to find a place to "die" in. Your corpse will be digested and decomposed by the environment, supporting it to flourish. After being recycled, you respawn, and get to do it all over again. You can visit previous places you have chosen to die in, and see how the environment responded to your corpse.*



"Climate Zoned: The Dating Simulator": *In a dating simulator, you get to meet and date different climate zones, presented as people, in a university environment. You can choose to romance and get to know each of them. Over the course of the game, you realize that the climate zones are changing due to climate change, and that ecological collapse is due soon.*

"The Witnesser": *In an adventure role-playing game, you take on the role of the "witnesser", a prestigious but grueling, sacrificial position that forces you to travel around to enable crop-growing and animal herding, as nothing non-human grows or moves if there is not a "witnesser" looking at it. As a player, you can choose to maintain the status quo, or investigate ways to break this tradition.*

"OzoooOOO000OOOooone": *In a turn-based strategy game, you take on the role of the hole in the ozone layer, as a sentient, self-aware being. As you move around, you can choose to communicate with humankind—or completely ignore it. You cannot save during the game, and each run of the game is final. As a player, you can choose to support humankind's interests, grow your own size and wreak havoc, or decide to not interfere with surface dwellers at all.*



REFLECTIONS

Canon/assumptions in human-nature relationships: This study did not develop as I had expected: Despite wanting to relate to the natural world more closely, the ways in which I could do so in the video games I played were limited. Here, a default current became visible: The human being as the one in power, ready to exploit, dominate, and shape nature in their image [15]. This portrayal of nature can be interpreted as echoing destructive colonial, and imperial views [20]. I struggled heavily with being given such an inflexible position, and it fostered a longing inside of me to engage differently with nature as a whole, and in games in particular. Through the sketches of video games, I explored a different set of interactions and narrative; in which the player becomes just one part of the world.

Different "verbs" for interactions with digital nature: My auto-ethnographic work prompted me to think carefully about how we portray non-humans in artistic mediums. My thinking here is very much informed by conversations that I had with other scholars in our university, that study how nature is portrayed in English literature: We are always operating, writing, drawing and making from a human-centered perspective. Even if we describe what a river might feel, what a tree's life is like or how a virtual landscape moves, we do so through an inherently human lens. Even if we are aware of this circumstance, we have almost no choice but to humanize—or even anthropomorphize—the non-human. I try to frame this circumstance as something neutral—it is what it is—because we encounter similar difficulties with other people as well: We can relate to one another, across a wide spectrum, but we can also never fully know what it is to be someone else. Here, we have modes of communication with other people to establish a shared baseline (e.g., language): We can correct our assumptions and adjust how we interact with each other, to bridge the gap between our existences. However, even with these tools available, people are prone to make mistakes, or to disregard them: To speak for others, to project onto them, and to disregard their agency. Engaging with the MtH means embracing that we can only ever approximate it; coming close, or closer, but never fully reaching it. We have to continuously wrestle with this circumstance.

DISCUSSION

The cases we discussed in this pictorial reflect ways of how we went about producing knowledge from the bottom-up rather than through theoretical engagement only, and as such they hint at the value of such an approach to practice-based MtH meaning-making. The different knowledges we produced leaned on active design practice as a key driver: they might not have emerged, or might have emerged differently, should we have approached them only theoretically.

An example is the methodological insights about attuning to the nature for understand whether there is a need for design, produced during Ferran's backpacking experiment: because they emerged from the bottom-up (that is, stemming from designing rather than from theoretical reasoning only), they are intrinsically tied to practice and, as such, immediately actionable. Similarly, Çağlar, by taking part in mushroom growth, discovered a turn-taking relationship with mushrooms in a design context that could be applied in other co-designing practices with the other-than-humans. In other cases, practice-based activities of imaginative role-taking and role-playing contributed insights into how decentering [29] could happen through action, by practicing it, to unpack the roles of (sometimes) unnoticed mediators in our experiences as actionable design insights for MtH design. For example, Eshtiak, by performing as robot dogs, was able to unpack the ways we expect a robot dog should behave and how it is mediated through the interfaces to control robots, as well as the robot's abilities, revealing the value of methods like "Wizard-of-Oz-ing" [26] for attuning to other-than-human perspectives. [Velvet]'s situated play activity helped them identify and provide insights into the ways to change human-centered values in the portrayal of nature in video games, while [Linas]'s embodied experience of walking in the city uncovered the unnoticed influences of non-organic elements like gravel and changing seasons as a lens for approaching city design.

Common in all of the cases we present, the knowledges didn't come from only a thorough engagement with the literature but from rich hands-on practices with other-than-humans. By not only engaging with 'what is' or 'what could be', but also and especially with 'what can designers do with/about it through embodied and situated actions/reflections', they exemplify the value of researching MtH design in a radically hands-on way, privileging action over sole theory. We believe that, as illustrated in our examples, the active engagement with design practice can give rise to new understandings that are both conceptually rich and practically actionable.

On the other hand, our emphasis on practice-based MtH design research is not an endorsement of relentless design practices disconnected from MtH values like care and kinship [40]. We recognize that designerly actions without such perspectives and reflections on our existing human-centeredness may contribute to the ecological and climate crises by adding more designs to those that caused these issues. Taking an MtH values-informed perspective that seeks co-existence rather than exploitation, we believe that our cases hint how such destructive and human-centered tendencies can be identified and altered through active designerly engagements: For instance, Ferran's case shows how a backpacking activity and attuning to other-than-humans with care can reveal when not to design, resisting the researcher's urge to create more technologies. Or, through playing video games with an expectation to feel closer to the nature portrayed in games, Velvet was able to identify human-centered values in game designs and presented alternative, caring, ways to engage with nature through play. Here, we emphasize our attempt to carry MtH values by reflecting on the mediators in our MtH design activities (Figure 1) helped us identify and adjust our potentially destructive practices in practices.

These examples highlight the value of active engagement with design practice when it comes to doing MtH design research. That was the primary purpose of our pictorial – highlighting and illustrating that value, hoping to support a move towards more designerly MtH design. Far from complete, though, that transition requires further research how to turn a key bottleneck of MtH approaches to address their actionability; or, in fact, the lack thereof. In that diretion, we see a need for more methods research that looks at how designers can engage with the MtH practically, actionably, responsibly, in a way that is not only critical and theoretically sound but also and especially implementable. To get there, we suggest design researchers should broaden perspectives and look far and wide beyond design research – we see a need for bringing MtH practices closer to actual design practitioners (those in the industry who arguably contributes to the ecological crises of our age, not us privileged thinkers in the sometimes self-contained bubble of academic research), to investigate how they perceive and engage (or not) with MtH ideals at the moment, and see what might be needed to build bridges they can rely and act on.

CONCLUSION

This paper highlights the critical importance of practice-based MtH design research, emphasizing the need to bridge the gap between theory and actionable design practices. Through five case studies, we demonstrated how hands-on engagement with design can yield valuable insights into MtH design. Each study provided unique perspectives on how non-human elements can actively shape design processes, from co-designing with natural environments to facilitating mushroom growth and role-playing non-human entities. These hands-on methodologies did not only enrich theoretical discussions but also provided concrete, actionable insights that can be readily implemented in design practice. By foregrounding embodied, situated, and designerly approaches, we encourage a shift towards a more practical understanding of MtH design.

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